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Bridge condition rating data modeling using deep learning algorithm

Heng Liu and Yunfeng Zhang

Department of Civil & Environmental Engineering, University of Maryland, College Park, MD, USA

ABSTRACT

This paper presents a deep learning-based bridge condition rating data modeling approach using selected data from the National Bridge Inventory (NBI) database. The objective of this research is to develop a data-driven approach that enables prediction of future conditions of highway bridge components from historical inspection data. The problem is solved by training a Convolutional Neural Network (CNN) model with online available NBI data. One prominent feature of the CNN model is that if well-trained it can represent the high dimensional data in the dataset abstractions for which conventional mathematical models may be difficult to describe. A case study of Maryland and Delaware highway bridges using historical data (1992–2017) sourced from the NBI database has been performed to demonstrate the proposed method. CNN models for three primary components of these highway bridges including the deck, superstructure, and substructure have been established. Optimization of model parameters is achieved through a parametric study. Research findings suggest that the deep learning model offers a promising tool as a data-driven condition forecasting approach for bridge components with a demonstrated prediction accuracy over 85%.

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1. Introduction

Bridge condition assessment is crucial to a successful Bridge Management System (BMS) as a proper measurement of bridge performance in communication with bridge engineers and administrations. Since 1972, the National Bridge Inventory (NBI) initiated by the Federal Highway Administration (FHWA) provides the primary source of data for bridge management in the United States (Chase, Small, & Nutakor, 1999; Frangopol, Kong, & Gharaibeh, 2001). According to the National Bridge Inspection Standards (NBIS), the highway bridges within the NBI shall be routinely inspected every two years and the condition ratings of bridge components (e.g., bridge deck, superstructure, and substructure) are recorded in the NBI database. The condition ratings are one-digit numbers that characterize the general condition of the entire component being rated in a scale from 0 to 9. A brief description on each scale of condition rating used in NBI are reproduced in Table 1 for completeness. Additional recording items and their descriptions can be found in the coding guide (FHWA, 1995).

A time-dependent reliability analysis requires the knowledge of deterioration models which predict the future structural condition and form the basis for life-cycle monitoring and maintenance planning. The condition ratings have been utilized as the basis of bridge component deterioration modeling to assist with the BMS in optimizing maintenance, rehabilitation, and replacement decisions. Existing approaches for deterioration modeling of bridge component conditions may be classified into three categories:

deterministic models, probabilistic models, and artificial intelligence (AI). Deterministic models usually describe the deterioration of bridge component conditions as a deterministic relationship in relation to selected independent (or explanatory) variables such as age, average daily traffic, and climate factors. The selection of these variables is usually based on engineering judgment and factor analysis (e.g., Morcou, Lounis, & Mirza, 2003; Goyal, Whelan, & Cavalline, 2017). The relationship may be formulated in the form of the polynomial or exponential/logistic (Bolukbasi, Mohammadi, & Arditi, 2004; Moomen, Qiao, Agbelie, Labi, & Sinha, 2016). As a result, identifying the set of influencing factors that can most adequately describe the deterioration process, and its best-fit functional form is time-consuming which typically needs prior engineering judgement with trial-and-error many times.

Probabilistic models assume the deterioration of condition ratings as a stochastic process to account for the inherent uncertainties and variations due to unobserved explanatory variables, the presence of measurement errors, and the inherent stochasticity of the deterioration process (Madanat, Mishalani, & Ibrahim, 1995). Markovian methods are reported as the most common probabilistic model and have been extensively studied in modeling such deterioration process (Cesare, Santamarina, Turkstra, & Vanmarcke, 1992; Frangopol, Kallen, & Van Noortwijk, 2004; Morcou, 2006; Ranjith, Setunge, Gravina, & Venkatesan, 2013). Markovian methods assume the bridge deterioration process as a Markov-Chain based stochastic process that describes the condition deterioration with a probabilistic

Table 1. Descriptions of condition ratings used in the National Bridge Inventory (FHWA, 1995).

Condition Rating	Condition	Description
9	Excellent	—
8	Very good	No problems noted
7	Good	Some minor problems
6	Satisfactory	Structural elements show some minor deterioration
5	Fair	All primary structural elements are sound but may have minor section loss, cracking, spalling, or scour
4	Poor	Advanced section loss, deterioration, spalling, or scour
3	Serious	Loss of section, deterioration, spalling or scour have seriously affected primary structural components. Local failures are possible. Fatigue cracks in steel or shear cracks in concrete maybe present
2	Critical	Advanced deterioration of primary structural elements. Fatigue cracks in steel or shear cracks in concrete may be present or scour may have removed substructure support. Unless closely monitored it may be necessary to close bridge until corrective action is taken
1	Imminent failure	Major deterioration or section loss present in critical structural components or obvious vertical or horizontal movement affecting structure stability. Bridge is closed to traffic but corrective action may put back in light service
0	Failed	Out of service – beyond corrective action

transition from one condition state to another during one inspection interval. The challenge of applying the Markovian models for bridge deterioration modeling includes the first-order Markov chain that implies the future condition depends only on the current condition state (Morcoux, Rivard, & Hanna, 2002; Frangopol et al., 2004).

As a generalization of Markov process models, Dynamic Bayesian networks (DBNs) have also been employed as probabilistic models for reliability-based deterioration modeling (Straub, 2009; Kosgodagan-Dalla Torre et al., 2017; Rafiq, Chryssanthopoulos, & Sathananthan, 2015; Yang & Frangopol, 2018). DBNs are a special class of BNs that are probabilistic models using directed acyclic graphs (DAG) to represent the joint probability mass function of a set of random variables, with time-variant variables being assumed with Markov property (Straub, 2009). The condition deterioration is modeled as a stochastic process with a set of model parameters, among which time-variant variables are usually assumed with Markov property. The DBN provides an important tool to probabilistically assess the condition and the reliability of structural elements at group levels, while the computation might become challenging with high-dimensional models or large number of random variables.

AI models aim to interpret the deterioration of bridge component conditions by capturing data abstractions from training dataset. The Artificial Neural Network (ANN) based methods are reported as the most common AI method (Sobanjo, 1997; Huang, 2010; Kawamura, Miyamoto, Frangopol, & Kimura, 2003; Li & Burgueño, 2010; Tokdemir, Ayvalik, & Mohammadi, 2000). For examples, Li and Burgueño (2010) conducted comparative studies of several ANN methods for predicting bridge abutment condition ratings in the state of Michigan and their models achieved an averaged prediction accuracy near 73% in identifying the true condition rating of damaged bridge abutments (with condition rating ≤ 4). Huang (2010) analyzed historical maintenance and inspection data of concrete decks in Wisconsin using ANN model with a reported accuracy of 75% in classifying the true condition rating.

Deep learning is a machine learning technique that allows computational models comprised of multiple processing layers to learn representation of data with multiple

levels of abstraction (Lecun et al., 2015). Among the deep learning algorithms, Convolutional Neural Network (CNN), have drawn growing interests in structural health monitoring (SHM) by leveraging on recent advancement in computer vision technology (e.g., Cha, Choi, & Büyüköztürk, 2017; Dorafshan, Thomas, & Maguire, 2018; Kumar, Abraham, Jahanshahi, Iseley, & Starr, 2018; Zhang, Rajan, & Story, 2019; Zhang et al., 2017). Encouraging results have also been reported in image-driven or vibration-based structural condition assessment applications (Rafiei & Adeli, 2018; Gao & Mosalam, 2018; Khodabandehlou, Pekcan, & Fadali, 2019; Liang, 2019; Liu & Zhang, 2019a, 2019b). For highway bridge applications, CNN has also been applied to process bridge inspection image data such as ground penetrating radar (GPR) and thermography for rebar localization or delamination detection in concrete bridge decks (Dinh, Gucunski, & Duong, 2018; Cheng, Shang, & Shen, 2019). One prominent feature of the CNN model is its ability to represent high dimensional data abstractions for which conventional mathematical model may struggle to describe. This motivates the present study of CNN for data mining of the NBI bridge inspection datasets.

This study investigates a CNN-based condition rating data modeling method for predicting condition rating of bridge components in future inspection using historical NBI data. The model can be used as a data-driven condition rating data modeling approach to assist with the decision-making process of road authorities. A case study of Maryland and Delaware highway bridges using historical data (1992–2017) sourced from the NBI database has been performed to demonstrate the proposed technique. CNN models for three primary components of Maryland highway bridges including the deck, superstructure, and substructure have been trained and validated. Optimization of model parameters has been achieved through a parametric study. Research findings suggest that the deep learning model offers a promising tool as a data-driven condition forecasting approach for bridge components with demonstrated prediction reliability. Bridge condition prediction results using the trained CNN models are also discussed to provide insights in the implementation.

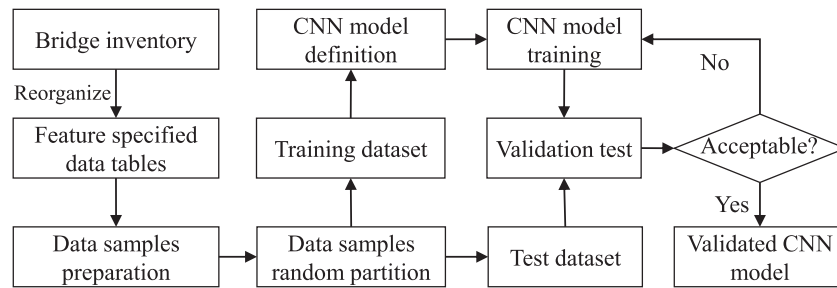


Figure 1. Flowchart of CNN-model based bridge component condition rating prediction method.

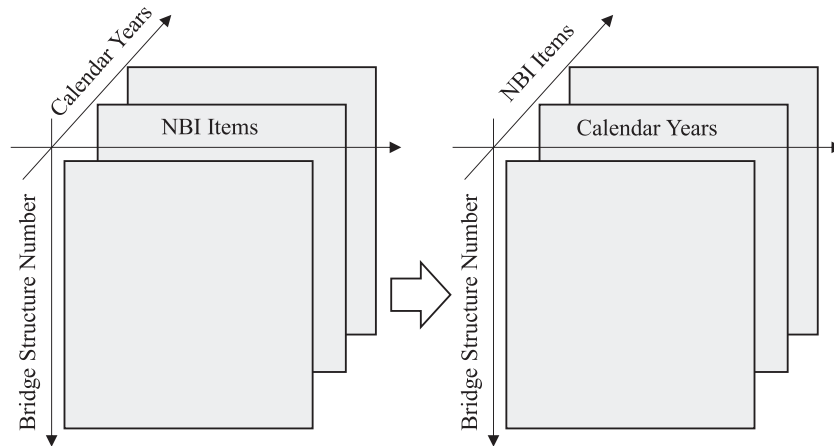


Figure 2. Reorganization of raw NBI database (left) into feature-specific data tables (right).

2. Deep learning based condition rating forecasting method for bridge components

2.1. Technical procedure

The proposed method for condition rating forecasting of bridge components utilizes a deep CNN model which can be trained with historical bridge inspection data such as those record items sourced from NBI. The bridge components here refer to the three primary bridge components: deck, superstructure, and substructure. The input data to the CNN model can be either the latest one or multiple historical inspection data including condition ratings, ages, and other features. The output is defined as the condition rating predicted for the immediate next inspection. The forecasting model that can predict the component's future condition based on its natural deterioration process or under current routine maintenance schedule is of the most interest, since such forecasts can form the basis of cost-effective decisions for major repairs, rehabilitation or replacement work.

To this end, an essential step of data pre-processing is performed to screen the data samples and filter out uninterested outliers. The screening step adopted here is intended to discard those data samples with condition improvement. Hence, the CNN model is trained to forecast the future condition of a bridge component by assuming there is no significant condition improvement work in near future while routine maintenance or minor repairs are still considered. Such prediction is thus based on the CNN model trained using processed data samples from historical NBI records and the model is optimized through an iteration process termed back-propagation. The accuracy of the bridge

condition forecasting derives from the deep learning model trained with large number of data samples that best represents the actual condition deterioration process. As a case study to showcase the process, the historical records of Maryland and Delaware highway bridges retrieved from the NBI database (1992-2017) are used here as the training dataset for the CNN model.

The technical procedure of the proposed bridge condition forecasting method is illustrated in Figure 1. First, the selected raw NBI database are reorganized into a data repository consists of feature specified data tables. The data table contains historical values of selected feature for each individual bridge, which are sourced from NBI record items. The reorganization is aimed at reducing the I/O time for preparing discrete data samples, since raw NBI files are compiled on an annual basis with mixed features (see Figure 2). Next, the reorganized data repository is prepared into discrete data samples consistent with the format requirements and properly labeled (more details in Section 3.1). Lastly, the prepared data samples are randomly assigned to the training dataset and testing dataset for optimizing and validating the CNN model. Once validated, the trained CNN model is ready for use by engineers for bridge condition deterioration modeling and future condition rating forecasting.

2.2. Deep learning model description

2.2.1. Convolutional neural network

CNN is a special type of artificial neural networks that performs the convolution operation in at least one of its layers. Examples of other computational layers in a CNN model

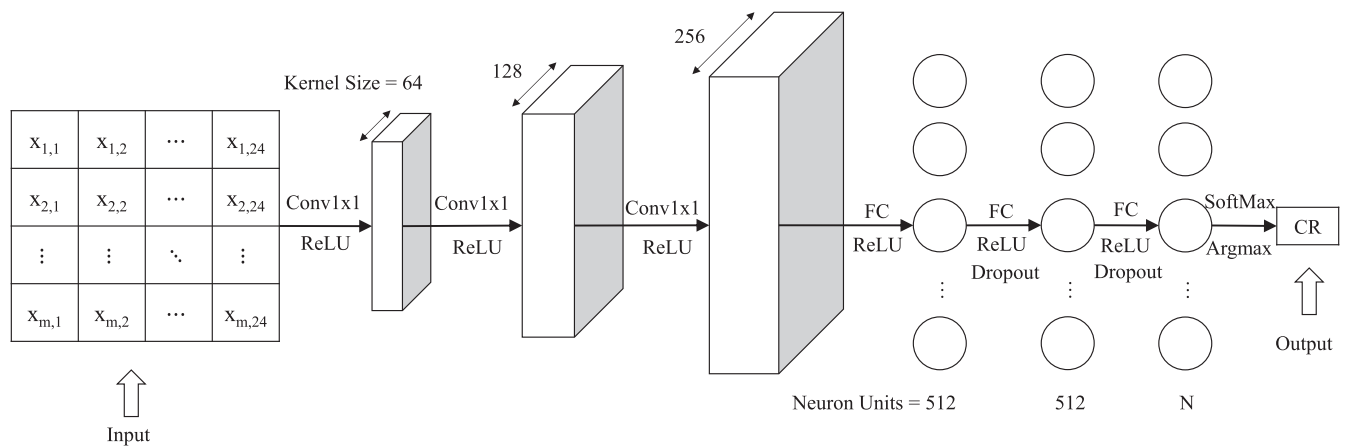


Figure 3. Architecture of the CNN model for bridge component condition rating prediction (FC: fully connected; CR: condition rating).

include the pooling layer, the rectified linear unit (ReLU) layer, and the fully connected (FC) layer. Details regarding the mathematical operation of these layers can be found in references (see, e.g., Goodfellow, Bengio, & Courville, 2016); Description of these layers is also given by Liu and Zhang (2019a). The design of the CNN architecture can be customized for specific task through experiments guided by monitoring the validation error, such as the deep CNN models reported with record-striking performance of image classification on the large-scale (order of millions) benchmark database of ImageNet (Deng et al., 2009), AlexNet (Krizhevsky, Sutskever, & Hinton, 2012), VGGNet (Simonyan & Zisserman, 2014), and ResNet (He, Zhang, Ren, & Sun, 2016). The common architecture of CNNs stacks a few convolution-ReLU layers, followed with a pooling layer, and repeats this pattern until it merges to the fully connected layers.

The structure of the CNN model adopted in this study is designed based on the common scheme, as illustrated in Figure 3. The features of the adopted CNN model are highlighted as follows:

1. 1x1 (filter size) convolution (Lin, Chen, & Yan, 2013) is employed throughout all the convolutional layers in the model to increase the representation power of the neural network as well as to simplify the architecture design process. A key aspect of designing the CNN filter (or the convolutional layer) is to capture the local connections by taking advantages of the properties of natural signals (LeCun et al., 2015). For example, in natural signals such as images, local groups of values (i.e., pixels) are often highly correlated, forming distinctive local motifs that can be easily detected (LeCun et al., 2015). In this study, the input data matrix is comprised of a list of selected features. Different features in the matrix are believed to be connected but may not be “local” to each other. For instance, the traffic volume of the bridge is related to the number of lanes and the GPS coordinates (i.e., location) of the bridge. However, the sequence of features (i.e., #1-2, #7-8; see Table 2) are sparse, and a large filter (e.g., 1x8) may be needed. Consequently, both the filter size

and the sequential arrangement of the selected features need to be considered, requiring extra work in the architecture design. The 1x1 convolution is thus adopted to simplify the design process. The connections between the features can be learned in the later stage of the network (i.e., fully connected layers). The number of filters (i.e., the kernel size) is designed by following the common scheme of those sophisticated CNN models such as VGG-16.

2. The pooling layer which is designed to reduce the spatial size of representations is discarded to maintain the simplicity of the structure. Springenberg, Dosovitskiy, Brox, and Riedmiller (2014) suggested that a homogeneous CNN solely consisting of convolutional layers can achieve the state-of-the-art performance without the need for max-pooling layers.
3. A dropout layer with a ratio of 0.5 (Srivastava, Hinton, Krizhevsky, Sutskever, & Salakhutdinov, 2014) is added to mitigate the overfitting effect.
4. The CNN model is trained in a classification scheme, and the number of neuron units in the last FC layer is set as N , which is equal to the number of possible condition ratings in classification. In this study, N is set to be seven by considering the possible condition ratings ranging from 3 to 9. The lowest condition rating considered in this study is 3 since very few samples have been found to have a condition rating number below 3 in the considered NBI dataset.

2.2.2. Cost function

The cost function is the objective function that defines the training target of the CNN model. In this study, the CNN model is treated as a classification model and the training target is to classify the future condition rating based on the given input. The output (i.e., future condition rating) of the CNN model are thus discrete classes. Since bridge component condition rating number is an integer number ranging from 0 to 9, it is natural to consider the condition forecasting as a classification problem with each class representing one possible condition rating number. The cost function for a classification model is usually formulated using the SoftMax with cross-entropy. The SoftMax function is used

Table 2. Features selected for deep learning model based bridge component condition forecasting.

Descriptive Information (Category)	#	Features (Influencing Factors)	Data Source from NBI
Geographic Region (A)	1	GPS Latitude Degrees	Item 16
	2	GPS Longitude Degrees	Item 17
Bridge Other Attributes (B)	3	Toll	Item 20
	4	Maintenance Responsibility	Item 21
	5	Functional Class of Inventory Rte.	Item 26
	6	Bridge Age	Item 27, 90, 106
	7	Lanes on Structure	Item 28A
	8	Average Daily Traffic (ADT)	Item 29
	9	Inspection Interval	Item 90
	10	Designated Inspection Frequency	Item 91
	11	Reconstruction	Item 106
	12	Average daily truck traffic (ADTT)	Item 109
Structure Configuration (C)	13	Skew	Item 34
	14	Structural Material/Design	Item 43A
	15	Type of Design and/or Construction	Item 43B
	16	Number of Spans in Main Unit	Item 45
	17	Length of Maximum Span	Item 48
	18	Structure Length	Item 49
	19	Bridge Roadway Width Curb-To-Curb	Item 51
	20	Deck Structure Type	Item 107
	21	Type of Wearing Surface	Item 108A
	22	Type of Membrane	Item 108B
Condition Rating (D)	23	Deck Protection	Item 108C
	24	Deck Condition Rating (CR)	Item 58
		or Superstructure CR	or Item 59
		or Substructure CR	or Item 60

here to represent the probability distribution over C different classes (Goodfellow et al., 2016), as described below:

$$P_{model}(\hat{y}_i|x_i, \theta) = \frac{1}{\sum_{i=1}^C \exp(z^i)} \begin{bmatrix} \exp(z^1) \\ \exp(z^2) \\ \vdots \\ \exp(z^C) \end{bmatrix} \quad (1)$$

where $P_{model}(\hat{y}_i|x_i, \theta)$ represents the conditional probability distribution over C different classes based on given input sample x_i and model parameters θ (including weights and biases); z^i ($i=1,2,\dots,C$) denotes the activations in the preceding layer (e.g. a fully connected layer). It is noted that the probability distribution is a vector with a length of C while the sum of this vector is equal to one. For instance, $P_{model}(\hat{y}_i|x_i, \theta) = [0.1, 0.1, 0.5, 0.2, 0.1]$ may be the output of a SoftMax function. The cross-entropy loss is used to measure the dissimilarity between the conditional probability distribution of the training dataset X consisting of n data samples and the distribution estimated by the CNN model as given by:

$$L(X, \theta) = -\frac{1}{n} \sum_{i=1}^n y_i \ln[P_{model}(\hat{y}_i|x_i, \theta)] \quad (2)$$

where y_i is the label of the data sample x_i and the $P_{model}(\hat{y}_i|x_i, \theta)$ can be represented using the SoftMax function in Equation (1). The label y_i is an encoding vector with a length of C , and the label for the i^{th} class is determined by assigning the i^{th} value of the vector as one while all others are zeros.

The loss function is taking the negative logarithm of the model estimated likelihood. Since $\ln(x) \in (-\infty, 0]$ for $\forall x \in [0, 1]$, the loss function is minimized towards zero. Minimizing the cross-entropy loss is equivalent to the maximum likelihood estimation of model parameters θ to match the data distributions (Goodfellow et al., 2016). From this

view point, the trained CNN model under classification scheme can be characterized as a probabilistic model based on learned data distribution. Once the CNN model is trained and validated, the condition rating forecasting for a given data sample (x_j) is made by choosing the class (i.e. condition rating) with the highest probability as expressed by:

$$\text{Forecasting} = \text{argmax}[P_{model}(\hat{y}_j|x_j, \theta)] \quad (3)$$

2.2.3. Training algorithm

The training of the CNN model is to search for the optimal model parameters θ that has the least cost function value. This can be achieved through an iteration process called gradient descent and its extensions. The generalized notation of the gradient descent may be described as:

$$\theta_{new} = \theta_{old} - \eta \frac{\partial L(X, \theta)}{\partial \theta} \quad (4)$$

where θ denotes each parameter including weight and biases in the CNN model, the derivative or gradient is calculated through the backpropagation method (Rumelhart, Hinton, & Williams, 1986), and the learning rate (η) is a user-specified parameter that defines the step size of each iteration. The value of the learning rate can be set in a variety ways. A common scheme is to set η initially as a relatively larger step (e.g., $\eta = 0.01$) to accelerate the training in the beginning and then periodically decrease the value to a lower number (e.g., $\eta = \eta \times 0.5$) to ensure that the step size is refined to achieve desired accuracy in the later stage.

One extension of the gradient descent for enhanced training performance is the stochastic gradient descent (SGD) with momentum. The term of stochastic means the calculation of the cost function $L(X, \theta)$, which can be

determined with Equation (2), is based on a randomly selected sub-dataset X rather than the entire training set. The sub-dataset X is called mini-batch, and the number of individual data samples in X is called the mini-batch size. The SGD with momentum introduces a velocity term in Equation (4) to guide the direction of parameter updating. The SGD with momentum is employed in this study as the training algorithm and details of the algorithm can be found in the reference by Goodfellow et al. (2016). The training progress can be monitored by recording the value of the cost function at each training iteration, and the training is considered to have converged if the cost curve reaches a value that cannot be further decreased, indicating the CNN model is not being further optimized. The training results of the CNN model reported in the following case study suggest that this algorithm has shown a satisfactory convergence performance.

2.2.4. Model performance measurement

To quantitatively evaluate the CNN model performance in forecasting, its performance metric (PM) is defined as the average prediction accuracy over samples with the current condition rating from 3 to 9, as follows:

$$PM = \frac{1}{k} \left(\frac{1}{7} \sum_{i=3}^9 \frac{n_i}{N_i} \right) \quad (5)$$

where n_i denotes the number of cases of correctly predicted samples with a current condition rating of i , and N_i is the total number of samples being predicted ($i = 3, \dots, 9$; a total of 7 possible condition ratings). The term k denotes that the performance metric is also averaged from k trained models using k -fold cross-validation. The k -fold cross-validation refers to a model validation technique that random partitions the original dataset to k subsets with equal size and employs $k-1$ of them for independent training of k models. Details about k -fold cross-validation may be found in other literatures (see, e.g., Bishop, 2006; Goodfellow et al., 2016). In this study, the k is set as 5. In addition to the performance metric, the output probabilities of trained CNN models are also assessed to provide insights of the trained CNN models.

3. Case study of Maryland and Delaware Highway Bridges

The proposed deep learning model based method is first used to build the condition forecasting models for the bridge deck component of Maryland and Delaware highway bridges. Bridge data from the two states are combined to augment the dataset size. The examination starts with a baseline model and follows with a parametric analysis for optimal prediction. The condition forecasting method is then demonstrated for the other two bridge components – superstructures and substructures.

3.1. Data preprocessing

First, the raw data files containing the Maryland and Delaware highway bridges in the NBI database were retrieved and reorganized into a data repository with the selected 24 features (or influencing factors; see Table 2 for details). It is noted that raw NBI database is organized per bridge structure number, NBI Item ID, and calendar year, which dramatically increases the I/O time for retrieving historical values of a specific feature. Thus, the reorganization, as illustrated in Figure 2, is performed by constructing data tables for selected features and each table has its row representing an individual bridge (distinguished by the structure number or Item 8 recorded in the data file submitted to FHWA in 2017) and its column representing the value history of the feature sourced from calendar year 1992–2017. For instance, there are 6,214 (5,335 in Maryland and 879 in Delaware) distinct structure numbers in the data file of 2017 and the data table has 6,214 rows $\times$ 27 columns with the first column as the structure number and the rest 26 columns as the corresponding feature value history from calendar year 1992 to–2017.

A cleansing procedure is also performed to discard redundant data points for each feature table. It is noted that the NBI records are published on a yearly basis while most bridges are inspected every two years except for the cases described in Item 91 (e.g., after major collisions or earthquakes), which results in redundant information in the retrieved data repository. For instance, a condition rating may be duplicated in two adjacent years (columns) in the data table marked by identical inspection date. Hence, each feature table is cleansed by only storing data points with distinct inspection date.

3.2. Feature selection

The choice of the selected 24 features are primarily made based on the influencing factors for bridge condition ratings observed in prior literature (Madanat et al., 1995; Chang, Maguire, & Sun, 2017; Ghonima, Schumacher, Unnikrishnan, & Fleischhacker, 2018; Goyal et al., 2017; Kim & Queiroz, 2017; Sanders & Zhang, 1994). For example, a faster deterioration was found for bridges in northern Indiana compared to those in southern Indiana, which was attributed to the use of deicing salts in cold weather regions (Mishalani & Madanat, 2002). The bridge geographical region, which is represented by the Global Positioning System (GPS) Latitude (Item 16) and Longitude (Item 17) coordinates in this study, may implicitly account for the influence of environmental factors such as the increased freeze-thaw cycles in colder regions.

As detailed in Table 2, most features are directly sourced from the NBI recording items; however, the bridge age is calculated by subtracting the year built (or the reconstruction year if applicable) from the inspection year, and reconstruction status is a binary number indicating whether a reconstruction has been implemented and recorded in Item 106 with 0/1 for No/Yes. The selected features are also grouped into 4 categories based on the similarity of their

Table 3. Dataset size when using the 24 features from the latest m inspection records.

Bridge Component	Total # of samples					
	$m = 1$	$m = 3$	$m = 5$	$m = 7$	$m = 10$	$m = 12$
Deck	39891	31192	23293	15781	5821	2470
Superstructure	40045	31336	23427	15864	5849	2482
Substructure	40159	31427	23482	15928	5908	2506

descriptive information. The grouping of the selected features is also aimed at reducing the number of possible permutations of feature combination in subsequent parametric study.

3.3. Dataset preparation

The organized data repository is thus prepared into discrete data samples consistent with the designated input data formats and being properly labeled. The label is defined as the actual condition rating of the bridge component in the immediate next inspection. The data sample uses the selected features with their values from the last m successive inspections ($m \geq 1$) to populate the input matrix for the CNN model. The matrix height represents the number of distinct inspection records (indicated by different inspection dates) and its width stands for the number of features considered. In the matrix, features in the same row are valued from the same inspection record, and the row sequence is organized with the most recent record at bottom and the earliest at top.

Therefore, by using a different setting of input matrix, the prediction by the well-trained CNN model either assumes that the future conditions are history-independent and depend only on the feature values in the current ($m = 1$) inspection record, or the future conditions are history-dependent and depends on the multiple ($m \geq 2$) feature records. Since very few data samples have a condition rating lower than 3, the condition rating of 0, 1, 2, and 3 are grouped together as 3 to indicate the bridge component is in failure condition requiring corrective action.

Data samples with at least one feature value of NA (not applicable) are discarded. Data samples with a label of condition rating greater than the rating in the current inspection year are also discarded since the current deterioration model does not consider significant rehabilitation or repair work by assuming that the predicted condition rating can only stay in its current value or drop to a lower value. The prepared data samples are then normalized by Z-score expressed in Equation (6) for each individual feature:

$$z_i = \frac{x_i - \mu_i}{\sigma_i} \quad (6)$$

where μ_i and σ_i are the mean value and standard deviation of corresponding feature x_i over the entire discrete sample dataset. The standardized dataset has a zero mean and a standard deviation equal to one. The normalization can help to speed up convergence, even when the features are not decorrelated (LeCun, Bottou, Orr, & Muller, 1998; Ioffe & Szegedy, 2015).

Finally, the normalized data samples are randomly partitioned into the training and testing datasets, with the

**Figure 4.** Sample learning curve of the CNN model.

training samples occupying 80% portion while the remaining 20% is allocated for the testing set. The total number of acquired data samples with considered matrix height (m) for the deck is listed in Table 3. It is noted that fewer data samples are available when multiple-inspection records are used to populate the input matrix.

3.4. Deep learning implementation

The Deep Learning Toolbox (2019) is employed to train and validate the CNN model with the architecture shown in Figure 3. Accelerated Graphics Processing Units (GPU) implementation is employed using an Nvidia GeForce Titan X GPU card. The stochastic gradient descent with moment is chosen as the solution algorithm, with a mini-batch size of 1,024 and a moment coefficient of 0.95. The learning rate (η) employs a piece-wise decayed setting that uses a relatively large value to accelerate the training in the beginning and the rate is periodically reduced by multiplying a decay factor to ensure that the learning rate is not too high when the training nears the optimal point.

In this study, the initial learning rate is 0.01 and the value is reduced after every 10 training epochs with a decay factor of 0.5. The total number of training iterations is set to be large enough so that the training reaches an optimum; in this study, the maximum training epoch is set as 30. The total number of training iterations is equal to the training dataset size divided by the mini-batch batch size and then times the maximum training epoch. Testing is performed after each training epoch. The convergence of well-trained deep CNN models can be identified by monitoring their learning curves with a stable low training loss that cannot be further reduced. Sample learning curve is shown in Figure 4.

3.5. Baseline model for Bridge Deck

The proposed method is first validated with a baseline model setup: using the 24 features (as listed in Table 2) of their values in the current inspection year (the size of input

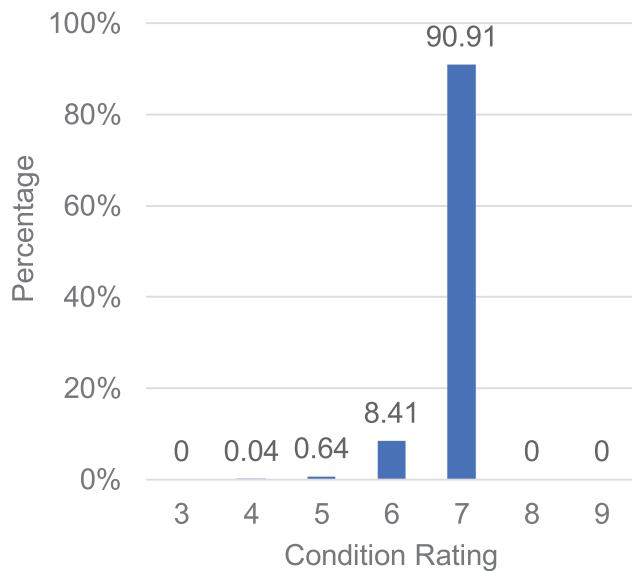


Figure 5. Empirical data distribution of future condition ratings for decks with a current condition rating of 7.

feature matrix is 1×24 ; see dataset size ($m = 1$) in Table 3) to predict the condition rating of the bridge deck in the next inspection cycle. The performance metric (PM) of the baseline model on the testing dataset is evaluated as 85.4% per Equation (5). The PM indicates the prediction capacity of the trained model on the independent testing datasets. The results indicate that the proposed deep learning model based method with the baseline model setup is fairly effective in condition forecasting for bridge decks, while a series of parametric studies are also done to provide insights for the model prediction.

3.6. Parametric analysis

3.6.1. Input matrix width

The input matrix width represents the number of input feature considered as the CNN model input. In practice, the contribution from different features to the condition deterioration of bridge components may vary in different ways; thus, selecting the set of best-fit features is time-consuming and may need trial-and-error of many times. One favorable character of the deep learning model is its automated feature-mining, and in theory the best set of most relevant features can be learned by itself with no need for their explicit definition. This aspect is examined by observing the model performance with varied input matrix width (i.e., different feature selections). The four feature categories described in Table 2 are permuted with eight feature combinations as listed in Table 4.

In addition to the baseline model (i.e., Model #1), seven additional CNN models are established by adapting the input layer to the number of the selected input features, while the rest model architecture is kept unchanged. For example, when the combination of geographical region and the condition rating (model #5) is examined, the input layer is modified from 1×24 to 1×3 , corresponding to the condition ratings, GPS latitude and longitude degrees. It is

Table 4. The eight CNN models and corresponding input features.

Model #	Feature Combination	Model #	Feature Combination
1	A + B + C + D	5	A + D
2	A + B + D	6	B + D
3	A + C + D	7	C + D
4	B + C + D	8	D

worth to mention that all the eight CNN models share the same dataset used by the baseline model, from which the corresponding data samples are adapted (or truncated) according to the input features.

The PMs of the eight CNN models on the testing datasets yield similar values near 85%, with the Model #1 being the best (85.4% of PM) and Models #5 and #8 being slightly worse (still 85.2% for PM), suggesting that these additional features are not significant contributor to the model prediction accuracy. However, further investigation on model output probabilities below shows that the eight models may not perform “equally” in terms of output probabilities even when they achieved very close PM values in prediction.

It is interesting to note that Model #8 which uses only a single condition rating as the input achieved fairly accurate predictions, which suggests the condition rating is a primary influencing factor to the predictor in this study. One possible reason is that condition ratings of bridge decks do not change within one inspection cycle in most inspection times. For example, in the processed dataset, there are respectively 100%, 96.5%, 94.0%, 92.1%, 90.9%, 69.4%, and 29.4% of deck inspection data samples remaining in the current condition rating (CCR) with a CCR value ranging from 3 to 9. The output probabilities from the trained CNN models are also analyzed, and the focus is given to the term that reflects the probability of decks to remain in the CCR within one inspection cycle. This term will be referred as “remain-in-CCR” in the following analysis. The output probabilities from Model #8 corresponding to the “remain-in-CCR” for decks with a CCR from 3 to 9 are found to closely match previously mentioned percentages.

According to Equation (3), the predictions are assigned to the class (condition rating) of the highest probability and a probability greater than 50% can guarantee that such class has the maximum likelihood. Recalling the training objective, the CNN model is trained to learn data distribution through maximum likelihood estimation. Since Model #8 uses one discrete variable as input, the data distribution can be empirically determined by calculating the percentages of samples belonging to each condition rating. For example, the empirical data distribution of future condition ratings for decks with a CCR of 7 is plotted in Figure 5. A close match can be found in Figure 6, which plots the distribution of output probabilities from Model #8 for decks with a CCR of 7, predicting either “remain-in-CCR” (Figure 6a) or dropping to the rating 6 (Figure 6b).

It is noted that the output probabilities from model #8 for decks with the same CCR are constants, which means decks with different features such as construction materials and geographical regions are assigned the same probability. As a result, the prediction from model #8 is inadequate for condition forecasting per individual bridge due to a lack of

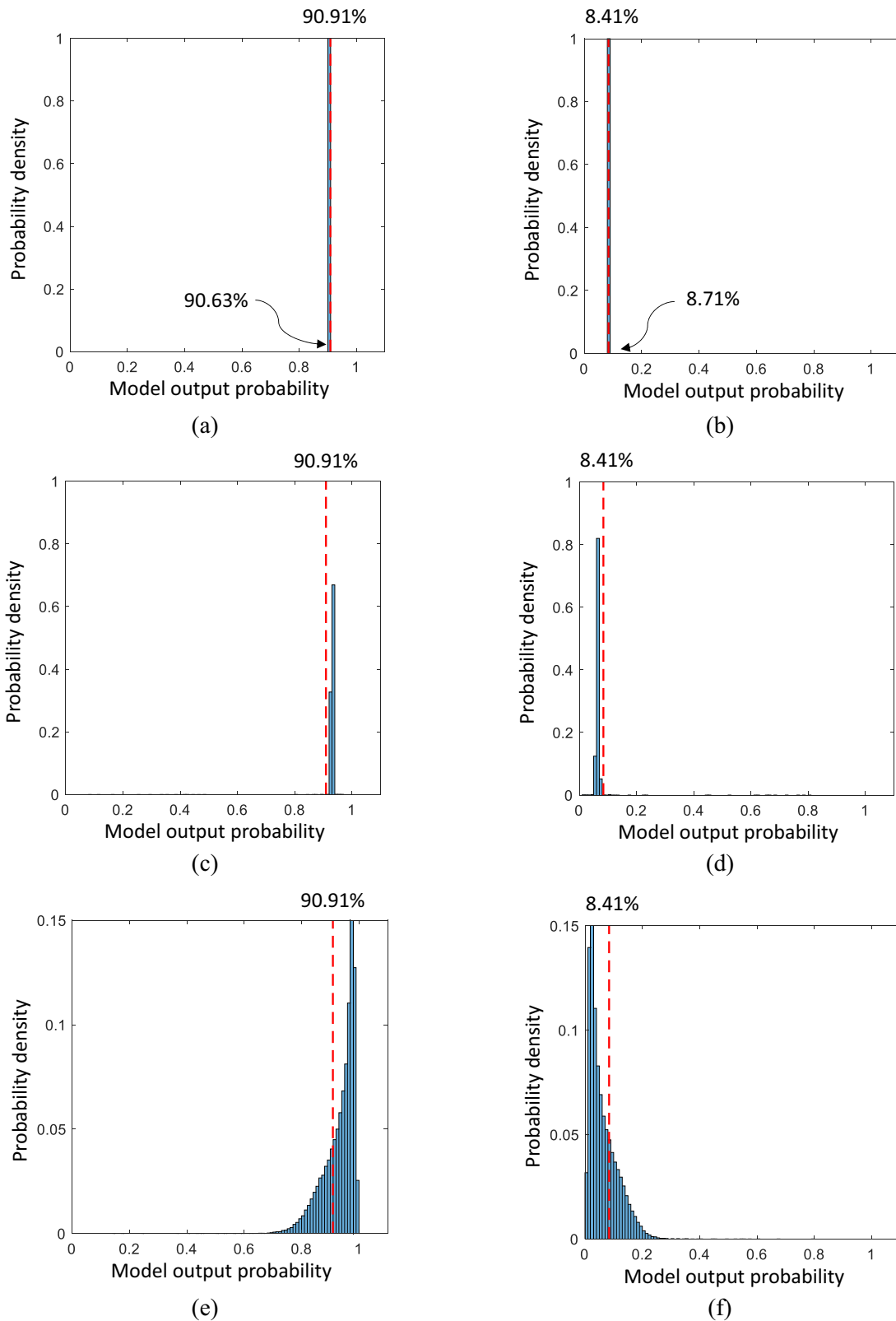


Figure 6. Model estimated probability distribution to remain in rating 7 (or drop to rating 6 for b, d, f) in one inspection cycle for decks with a current condition rating of 7: (a) Model #8, (c) Model #5, and (e) Model #1; drop to rating 6: (b) Model #8, (d) Model #5, and (f) Model #1.

resolution on bridge features. As a side-by-side comparison, the histograms of output probabilities from Models #1 and #5 are also analyzed, with the same instances plotted in Figure 6. Comparing to Model #8, slightly wider distributions are observed in Model #5 with two additional features

of GPS coordinates. The wider dispersion indicates that the geographic region does have influences the bridge deck condition deterioration.

On the other hand, Model #1 with 24 input features displays noticeably wider distribution than Models #5 and #8,

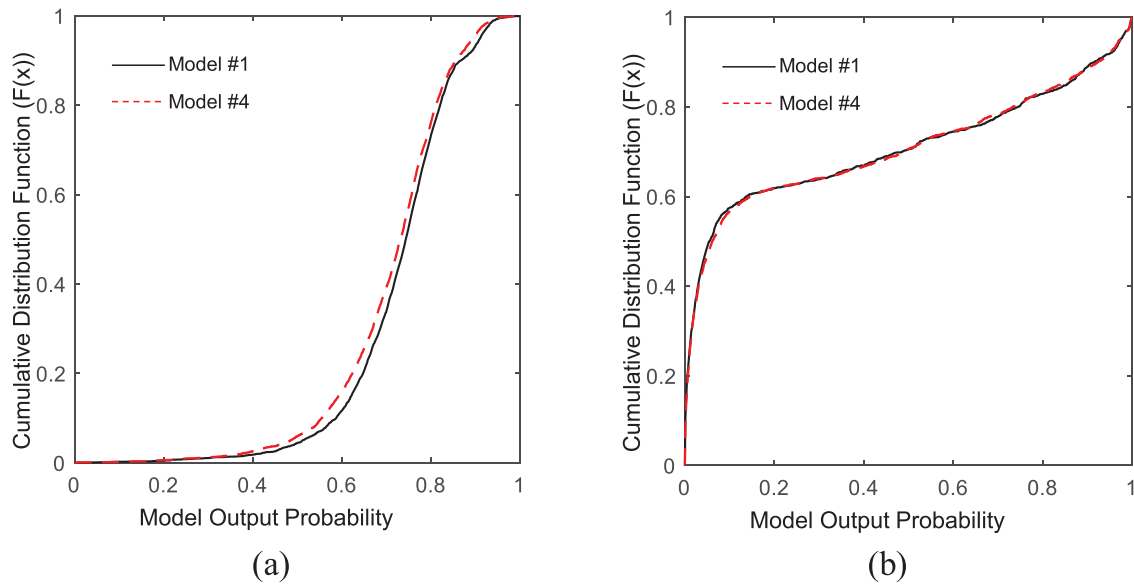


Figure 7. Comparison of empirical cumulative distribution functions of the model output probability remaining in current condition rating (CCR) for bridge decks with the CCR of: (a) 8; (b) 9.

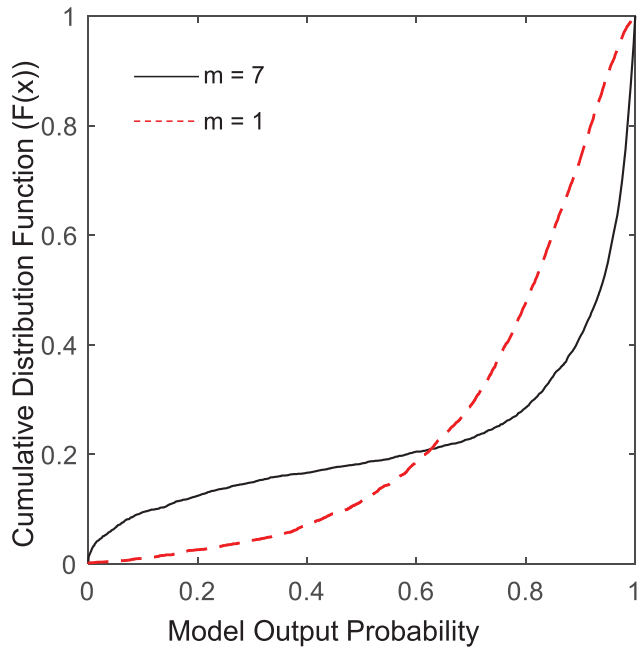


Figure 8. Matrix height effect on the cumulative distribution functions of the model output probability remaining in current condition rating (CCR) for bridge decks with the CCR of 8.

indicating that the number of input bridge features do affect model output probabilities. While expanding input matrix by adding more features such as the GPS coordinates has been found to have insignificant influence on the model prediction accuracy, the distributions of model output probability in Figure 6 reveal that additional features do have some effect on the model outputs. This finding indicates two models with prediction accuracies close to each other do not necessarily mean that two models are equal, since their predictive probabilities may be quite different. Consequently, the eight models may not perform “equally” in terms of output probabilities even when they achieved very close PM values in prediction.

The two-sample Kolmogorov–Smirnov (KS) test is introduced here to offer a nonparametric mathematical tool that statistically examines whether the two probability distributions are the same (Kolmogorov, 1933; Smirnov, 1948). The KS test measures the maximum absolute difference between the two cumulative distribution functions as follows:

$$D^* = \max(|F_1(x) - F_2(x)|) \quad (7)$$

where $F_1(x)$ and $F_2(x)$ are the proportion of sample values less than or equal to x .

When two models have been trained using the same procedure except that one model does not include the weak feature in the input matrix, the null hypothesis (H_0) is that if the weak feature does not influence the model prediction, the output probabilities from the two models shall be statistically identical. The null hypothesis would be rejected if the p -value is smaller than the significance level of α as:

$$\Pr(D > D^* | H_0) < \alpha \quad (8)$$

The p -value is calculated using the method by Marsaglia, Tsang, and Wang (2003). Since observed deck samples will most likely be assigned the same rating value as the CCR, the output probabilities corresponding to “remain-in-CCR” from the two models #1 and #4 are tested with the KS statistics; and the results show that the null hypotheses are rejected at a significance level of 0.05 for decks with a CCR from 4 to 8. For illustration, the discrepancy between the empirical cumulative distribution functions of the two model outputs for decks with a CCR of 8 are shown in Figure 7(a), and the KS statistic D^* is calculated as 0.054 with the p -value of $2.2\text{E-}4 (<0.05, \text{rejected})$.

Additionally, the converged empirical cumulative distribution function for decks with a CCR of 9 from the two models are also shown in Figure 7(b), and the KS statistic D^* is calculated as 0.027 with the p -value of 0.69 ($>0.05, \text{accepted}$). This finding suggests that the GPS coordinates statistically influence the model predictions in terms of

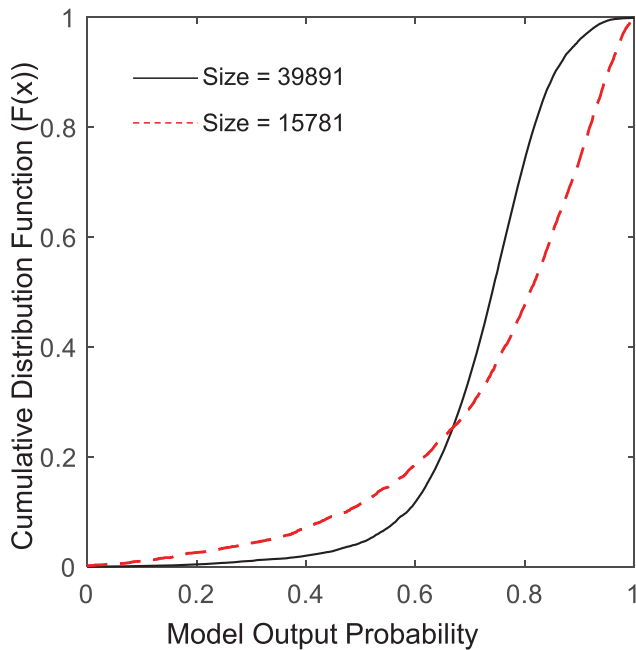


Figure 9. Dataset size effect on the cumulative distribution functions of the model output probability remaining in current condition rating (CCR) for bridge decks with the CCR of 8.

model output probability for decks with a CCR from 4 to 8 while have minor effect on decks with a CCR of 9. The converged cumulative distribution function from the two models also demonstrates that the CNN model can learn to “ignore” minor features that do not significantly contribute to the condition rating forecasting.

3.6.2. Input matrix height effect

The matrix height m denotes the number of past inspection records used for prediction. The total numbers of available data samples in Table 3 show that the dataset size shrinks to below 5000 when m is equal or greater than 10, and thus considered values of m in this study are 1, 3, 5 and 7. The examination is to investigate how the performance of trained CNN model varies with different settings of m . The comparisons are made based on prepared datasets containing identical information from the current inspection while additional data information from preceding inspections vary with different setting of m . This is achieved by first building the control dataset with m equal to 7. When preparing dataset for other values of m ($m=1,3,5$), the control dataset is first copied and each sample is truncated by keeping the corresponding rows. This implementation can also maintain prepared datasets with the same size, which may potentially affect the model predictions as discussed later.

Herein, the model with m equal to 1 is marked as Model #9 to distinguish from model #1 since their dataset sizes are different. The evaluated PMs are respective 85.0%, 85.0%, 85.8%, 85.9% for models with m being 1, 3, 5, and 7, which indicates that the matrix height marginally affects the model prediction accuracy. A subsequent investigation is then conducted by comparing model output probabilities between the two with m being 1 and 7. The focus is given to the term regarding “remain-in-CCR” and the KS test results

show that the null hypotheses are rejected at a significance level of 0.05 for decks with a CCR from 4 to 9. As an example, the discrepancy on the empirical cumulative distribution function from the two models for decks with a CCR of 8 can be seen in Figure 8.

3.6.3. Dataset size effect

It is noted that the PM of model #9 is slightly different from model #1 reported earlier, while the training procedure of two models is identical except for their respective dataset size (model #1 has a size nearly 2.5 times larger). This motivates the investigation on the dataset size effect on the model prediction. The distributions of model output probability regarding “remain-in-CCR” from the two models are thus compared with the KS test. The results show that, with the setting of m being 1, model predictions on bridge decks with a CCR from 4 to 9 are statistically affected by the dataset size (the null hypotheses are rejected at a significance level of 0.05).

As an example, the discrepancy on the empirical cumulative distribution function from the two models for bridge decks with a CCR of 8 is shown in Figure 9. The finding indicates that the model predicted data distribution may vary with increased dataset size. As a result, a re-training may be recommended once large dataset becomes available. Since this study only uses the Maryland and Delaware state highway bridge data, the findings (e.g., history-dependency and performance improvement) from the study of input matrix height effect are worth further study using expanded dataset in the future.

3.7. Condition rating forecasting of bridge superstructure and substructure

The proposed method is also validated on different bridge components. Similar procedures are adopted to prepare data samples, except that Feature #24 uses corresponding condition ratings such as Item 59 (superstructure) and Item 60 (substructure). The processed datasets of these two bridge components, with different setting of m , have similar size comparing to the bridge deck in Table 3. Considering the auto-feature mining capability of the CNN model, this validation study is based on the models using all 24 features.

Similar to the bridge deck, the evaluated PMs increase from 86.8% to 87.3% and from 85.1% to 85.4% for the respective superstructure and substructures when m increases from 1 to 7. It is noted that the comparison is made with the control dataset ($m=7$) prepared using the method described in Section 3.6.2. Quick scanning of data samples shows similar percentage patterns as the bridge deck that most of bridge superstructure and substructure stayed in the same condition in one inspection cycle; thus, the model output probabilities from trained CNN models are also analyzed with the KS test.

The test results show that model predictions on both superstructures and substructures with a CCR from 4 to 9 are also affected by the input matrix height m (the null

hypotheses are rejected at a significance level of 0.05). This may suggest that the future conditions of bridge superstructure and substructure are history dependent. However, further study is needed using expanded dataset since the dataset size also shows effects for model predictions on these two bridge components with a CCR from 4 to 9 (the null hypotheses are rejected at a significance level of 0.05).

3.8. Discussion of the results

The CNN algorithm provides a data-driven method for bridge condition rating forecasting which has the potential of adaptive model updating and is capable of individual bridge case prediction (vs. group level for most probabilistic or stochastic methods). Comparing to existing methods, the advantage of the proposed method is the auto-feature mining capability of the CNN model, which enables the model to automatically learn representations from training data of adequate size. Such advantage has been found to be very convenient in processing bridge inspection data, in addition to images or vibration signals reported in the references. In view of other promising results reported in machine learning fields (e.g., LeCun et al., 2015), it is envisioned that the deep learning models will find more applications in civil infrastructure condition assessment and deterioration modeling with unsupervised training and more available data.

Regarding its limitations, large amount of data is usually desired to ensure successful implementation of CNN model while the data quality is also critical due to its data-driven model nature. For example, condition rating data only indicates a general condition of rated component based on visual inspection which might introduce subjective error to the data (Phares, Rolander, Graybeal, & Washer, 2001); thus, the trained CNN model in its present form only reflects the inherent relationships in the training data and more research is needed for advanced data processing and objective data collection such as nondestructive evaluation (NDE). Data fusion with NDE data is desired to further improve the prediction accuracy.

4. Conclusions

A deep learning model based approach is developed for condition rating forecasting of three types of bridge components including bridge deck, superstructure and substructure. Convolutional neural network (CNN) is adopted as the deep learning model and trained for bridge condition forecasting in the present study. A case study using NBI data of Maryland and Delaware highway bridges is carried out to demonstrate the proposed method. Specific findings are summarized as follows:

- The CNN model can be trained to represent the high-dimensional data abstractions such as multi-factors and history-dependent effects on bridge components condition deterioration. The trained CNN models can achieve prediction accuracies over 85% on the independent testing datasets.
- Considering the auto-feature mining capability of the CNN model, the representation using all 24 selected features is believed to achieve the optimal predictions with adequate dataset size. For bridge decks, the evaluated PMs are 85.0%, 85.0%, 85.8%, 85.9% for models with m being 1, 3, 5, and 7 respectively, which indicates that the matrix height marginally affects the model prediction accuracy. Similar to the bridge deck, the evaluated PMs increase from 86.8% to 87.3% and from 85.1% to 85.4% for the respective superstructure and substructures when m increases from 1 to 7.
- Future conditions of the three types of bridge primary components have been found to be history dependent with the processed dataset, evidenced by the rejection of the null hypothesis at a significance level of 0.05. However, the history-dependency needs further study with expanded database since dataset size has also been shown to have some effects on prediction results.

Disclosure statement

No potential conflict of interest was reported by the authors.

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